7.6.2 Cultural Characteristics

- They are aerobic and facultative anaerobe. Optimum temperature is 37°C and pH 7.2.
- They grow on the following media and show the characteristic colony morphology (Table 7.5).

Table 7.5: Colony Morphology of *Corynebacterium diphtheriae* on cultural media

Media	Colony Morphology
Loeffler's Serum slope	They grow on this medium very rapidly. Colonies appear after 6–8 hours of incubation. The colonies are small, circular white or creamy and glistening
Tellurite Blood Agar	Grey or black colonies. Based on colony morphology on tellurite medium, three main biotypes – Gravis, Intermedius and Mitis.

Toxin

- The pathogenicity is due to production of a very powerful exotoxin by virulent strains of diphtheria bacilli.
- The toxigenicity of diphtheria bacillus depends on the presence of a tox^+ gene which can be transferred from one bacterium to another by lysogenic bacteriophages, of which beta phage is the most important.

Properties

The diphtheria toxin is a heat – labile protein and has a molecular weight of about 62,000 Dalton. It consists of two fragments.

a. Fragment A (24,000 Dalton) – It has all enzymatic activity.
b. Fragment B (38,000 Dalton) – It is responsible for binding the toxin to the target cells.

Mode of Action

The toxin acts by inhibiting protein synthesis, specifically fragment A inhibits polypeptide chain elongation in the presence of NAD by inactivating the elongation factor (EF – 2) the toxin has special affinity for myocardium, adrenal gland and nerve endings.

7.6.3 Pathogenicity

Source of infection – Airborne droplets

Route of entry – Upper respiratory tract

Incubation period – 3–4 days

Site of infection – Faucial (nasal, otitis, conjunctival, laryngeal, genital) diphtheria is most commonly seen in children of 2–10years.

Faucial diphtheria is the most common type. The infection is confined to humans only. The toxin has both local (flowchart 7.3) as well as systemic effects.

Systemic effects

The toxin diffuses into the blood stream and causes toxemia. It has got affinity for cardiac muscle, adrenal and nerve endings. It acts on the cells of these tissues.

7.6.4 Clinical Manifestations

1. Laryngeal obstruction, asphyxia (it is a condition of severe deficient supply of oxygen, causing suffocation).
2. Diphtheritic myocarditis (inflammation of heart muscle), polyneuropathy

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